DT081A - Signal and Image Processing

Lab 1 Report

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1 Introduction

[Briefly describe the lab objectives and the main concepts explored. Provide context for the experiments performed.]

2 Experimental Setup

[Outline the methods and tools used in the lab. Include any specific parameters or settings that were important for the experiments.]

3 Results and Analysis

3.1 Signal Generation and Visualization

1 (a) The sawtooth signal can be mathematically represented by the function:

$$x(t) = 2((ft + 0.5) \bmod 1) - 1.$$

Multiplying f by t gives us how many cycles have passed at time t. Adding 0.5 shifts the signal by half a period so that it is centered around $t = 0$. The modulo ensures the value within the interval [0, 1) and makes the signal restart after each period. Then multiplying by 2 and subtracting 1 scales the signal to the range [-1, 1).

1 (b) The frequency f is defined as the number of oscillations per unit time. When the frequency is doubled from 10 to 20 Hz, each cycle takes half as much time, which makes the signal appear more compressed in the time direction.

1 (c) Modifying the square wave still alternates between -1 and 1, but unlike before it starts at -1. This is because $np.ceil(0)$ is 0, and after multiplying by 2 and subtracting 1, the result becomes -1 and positive sine values returns 1.

1 (d) The composite signal consists is the sum of the sine wave and the square wave. When the square wave is at +1, the sine wave is shifted up, and the square wave is at -1, it is shifted down. This results in a signal that oscillates between -2 and 2.

1 (e) It can help to determine how a signal should be processed. For example, in audio processing, an equalizer can increase or decrease different frequency ranges such as bass, midrange and treble.

3.2 Fourier Transform and Frequency Domain Analysis

2 (a) The frequencies of the signal are 10 Hz and 20 Hz. They can be identified in the plot by the two peaks that are located at approximately 10 and 20 Hz.

2 (b) Adding a 30 Hz component introduces a new peak in the frequency domain plot. Since the new component has the same amplitude as the 10 Hz, their peaks are of equal are approximately the same height.

2 (c) The original signal contains strong, specific frequency components at 10, 20 and 30 Hz, while the random nois is spread across many frequencies instead of being concentrated at one spcific frequency.

2 (d) The chirp signal i created starts at a low frequency and gradually increases over time, which makes the oscillations become closer together because more cycles occur within the same time frame. In the frequency domain, the signal is spread over approximately 5-50 Hz because the chirp passes through many different frequencies, which is the reason why the plot shows a frequency range instead of a single peak.

2(e) The frequency domain can be used to see which frequencies range a signal occupies, which is important because wireless systems operate in specific parts of the electromagnetic spectrum.

3.3 Sampling and Aliasing

3(a) The Nyquist rate is 20 Hz and 15 Hz is not sufficent enough.

3(b) Increasing the sampling rate to 25 Hz gives it more samples to be above the Nyquist rate of 20 Hz, so the values contain enough to represent and reconstruct the original signal.

3(c) The signal is sampled at 30 Hz, which is below the Nyquist rate of 40 Hz. Therefore aliasing occurs. The 20 Hz signal appears as a lower frequency component at 10 Hz because frequencies above half the sampling rate are folded into lower frequency range.

3(d) The original signal has a frequency of 25 Hz and it's being sampled at 12 Hz. Because the sampling rate is lower then the expected Nyquist rate, aliasing occurs. This makes the signal appear to have frequency at 1 Hz, even though the orignal signal has 25 cycles per second.

3 (e) A scenario where aliasing is important is when an audio signal is sampled at a rate that is too low. High frequency components can then be reconstructed as incorrect lower frequency. Therefore, the sampling rate must be above the Nyquist rate in order correctly capture the highest frequency in the signal.

3.4 Image Sampling and Quantization

4 (a) As the resampling factor increases, fewer pixels from the original image are kept,which makes it less detailed. At factor 3, the distortion becomes more apparent, while at factor 4 it is clearly visible.

3.5 PSNR Calculation

5 (a) According to the PSNR values, the resampled image gave the lowest score with 22.71 dB, compared to 29.21 dB for the requantized image. A lower PSNR means the image is further from the original, so resampling introduces more distortion.

5 (b) The PSNR values agree with what i see, because the resampled image looks worse than the requantized one. But when i look at the image, my own aesthetic judgement mixes with the PSNR value, which makes me think the resampled value should be lower, maybe around 18-19 dB. PSNR only measures the pixel difference from the original, so it does not see the image the way a human does.

5 (c) In Figure 1, we can see tha the PSNR goes down drastically when the factor goes from 2 to 4, from 28.30 to 22.71 dB, which shows that the image quality drops a lot even with a small increase in factor. In Figure 2, the PSNR values show an almost linear relationship with bpp, which means that more bits per pixel gives a higher PSNR and better quality.

5 (d) PSNR is simple to calculate, has a clear physical meaning and is mathematically conventient, but it does match the perceived visual quality very well. SSIM instead compares three things between the original image, luminance, contrast and structure. This is based on the idea that the human visual system is adapted to extact structural information, which PSNR does not take into account [1]

5 (e) Both metrics agrees that requantization gives better results. The PSNR is 29.21 dB for requantization and 22.71 dB for resampling, and the SSIM is 0.88 for requantization and 0.66 for resampled, where 1 means identical to the original image. PSNR compares the images pixel by pixel, which gives a simple mathematical measure. SSIM instead compares the structure of the images, and the values show that the resampled image loses its structure because the edges become blocky, while the reqauntized keeps its structure but has fewer grey levels.

3.6 Noise Addition and PSNR Analysis

6 (a) I first started to see the noise at a standard deviation of 9, when i glanced back and forth between images and noticed grainy pixels on the smooth skin of the woman (Figure 3). This corresponds to a PSNR value of 29.04 dB.

6 (b) The cited threshold agrees with my result, since i started to see the noise at 29.04 dB, just shy of 30 dB. This makes 30 dB a good estimation of the threshold for acceptable image quality.

6 (c) Within the given interval 0-50, the noisy image was still recognizable, but at a deviation of 460 it became completely unrecognizable, as shown in Figure 4. The PSNR value at that deviation is 6.32 dB.

6 (d) To remove the noise from the image with a deviation of 35, i used a 3x3 median filter, which increased the value from 17.45 to 24.45 db, an increase of 7 dB. However, it did not reach the acceptable quality threshold disscussed in 6 (b). The PSNR value also does not show that details in the image are lost, because the filter removes so called pixel outliers. This can be seen in Figure 5.

6 (e) A more advanced method to remove noise is Non-Local Means. A medium filter only looks at the area around each pixel, but Non-Local Means searches whole image for patches that look alike and average them togheter. Because of this, noise can be removed without blurring away as many details, which was the problem with the median filter in Figure 5. Today, deep learning

methods using neural networks perform even better, as they learn what clean images look like from large amounts of data [2].

3.7 Audio Signal Analysis

7 (a) The audio clip is 502 miliseconds long, as shown in Figure 6. The highest frequency component is 4833 Hz. This is due to the fact that the weaker audio frequencies that are lower than 1% of the strongest frequency were disregarded, since otherwise even small noise peaks would have been counted.

7 (b) The bright horizontal lines under 5 kHz in Figure 7 represents the fundamental frequency and its overtones, since they are the strongest and the brightest.

7 (c) The FFT sorts the signal into frequency bins that match its frequency strength, and as seen in Figure 8, the peak of the fundamental frequency becomes larger and goes from 2400 to 3000 Hz, while its position stays around 220 Hz.

7 (d) Different audio files of speech, music and a machine sound were chosen, with different time lengths and different sounds, which gives different outputs, as shown in Figure 9. The electric shaver keeps the same strength through the whole 22s without any pauses, and its spectrum is spread over many frequencies up to roughly 6 kHz, with the strongest peaks reaching 500 and 700 Hz. Since the sound never changes, the spectrogram looks the same from start to end. The music starts quiter, because only the drums play first 10s before the bass and other instruments come in, and each drum hit shows up in the spectogram as small vertical lines. The speech is only 0,701s long and has two bursts with short silence between them, one for each word, which also appear as two seperate bright patches in the spectogram.

7 (e) A downsample function was implemented to reduce the sample rate, using the factors 1,4,6 and 8. Factor 1 is the original and will be used as the comparison. At factor 4(11025 Hz) there is only a small difference, but the sound feels slighlty grainy and muffled and seems to end more abruptly. At factor 6 (7350 Hz) the sound becomes muffled and slightly distored, since higher overtones are lost. At factor 8 (5512 Hz) the sound becomes very dull and distant, as mostly the low frequency remain. The first small extra peaks between the harmonics appear at factor 6, which is aliasing.

7 (f) A common processing technique is pitch shifting, where the pitch of the sound is shifted to be either higher or lower without changing the length of the audio. It can be done with resampling, as in 7 (e), combined with dividing the audio into smaller pieces like in the spectogram in the question before, and then stretching them out in time to match the original sound. The result is a sound with a different pitch but same length. This method is called a phase vocoder. It is used in the music industry, where autotune and other voice filters are used.

4 Discussion

[Interpret the results in the context of signal and image processing concepts. Address any unexpected findings or challenges encountered during the lab.]

5 Conclusion

[Summarize the key findings and insights gained from the lab. Reflect on how this lab contributed to your understanding of signal and image processing concepts.]

Appendix: Code Snippets

[Include key code segments used in the lab. Provide brief explanations for each snippet to demonstrate understanding.]

References

[1] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, Apr. 2004. DOI: 10.1109/TIP.2003.819861.

[2] M. Elad, B. Kawar, and G. Vaksman, *Image Denoising: The Deep Learning Revolution and Beyond – A Survey Paper –*, Jan. 2023. DOI: 10.48550/arXiv.2301.03362. arXiv: 2301.03362 [eess.IV].